

Autonomous book retrieval on TIAGo Pro

Emirates Robotics Competition 2026

Phase 1 — Simulation

Team Kanban

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Repository

github.com/4waiz/Rafeeq-library

Validated against

organiser dfl-rlab/erc_sim_2026 @ 2aa5a4a (v1.0.3-1-g2aa5a4a)

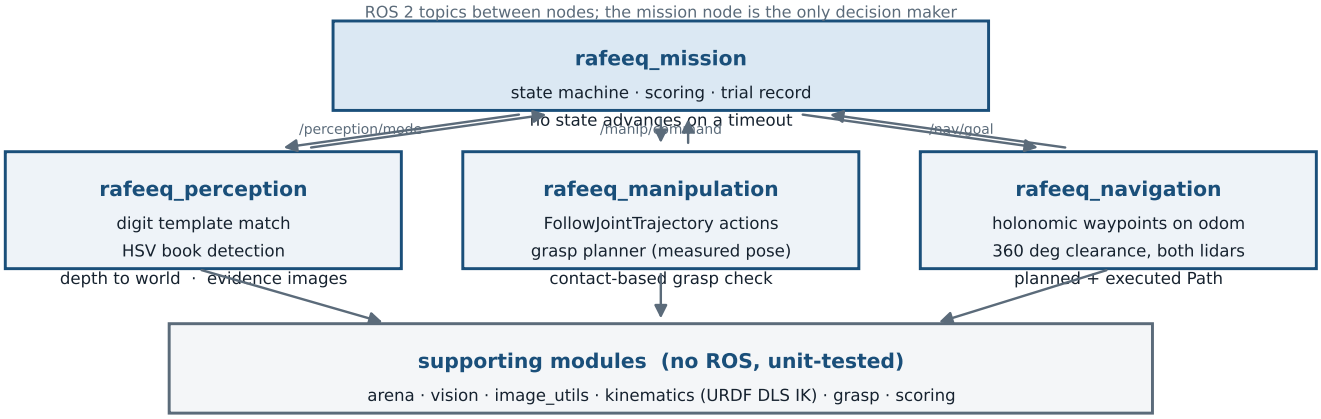
Ubuntu 22.04 · ROS 2 Humble · Gazebo Sim 8.15.0 (Harmonic)

Task completed end to end on the live simulator: book_col_3_row_2_red in the collection bin at t+107.8 s, reported by the organisers' own /bin_contacts sensor. One unedited take, 4:34.

Five validation trials, their rates, and the limits of them: sections 5 and 6.

1 System architecture

Rafeeq is four ROS 2 nodes plus six supporting modules that carry no ROS dependency and are unit-tested offline. The mission node is the only component that decides anything: perception, navigation and manipulation each expose a narrow command-and-status contract and never talk to each other. That separation is what made the defects in this system findable at all.



Interfaces, in the same order: mission on /rafeeq/*, /bin_contacts and /contacts; perception on RGB-D, CameraInfo, TF and /erc/*; navigation on /cmd_vel, /odom and /scan_*_raw; manipulation on FollowJointTrajectory actions and /joint_states.

Perception to action, end to end

The mission asks perception for a column; perception reads the number plates, deprojects each into the world, and publishes the physical column together with an annotated image. The mission drives there and asks for the book; perception returns the row and the measured 3D position. The mission passes that point to manipulation, which plans a grasp posture against it, and drives the base until the book is square in front of the arm. Nothing in the chain uses a nominal column centre: the organiser jitters each book by up to 0.25 m, three times the gripper opening.

What is deliberately not used

- Nav2 is installed but never brought up by the organiser launch: no map server, no AMCL, no costmap, no map frame. Rafeeq drives odometry waypoints and publishes the intended waypoint path and the executed trajectory as nav_msgs/Path, labelled as such rather than as a Nav2 plan.
- MoveIt is installed but has nothing to plan against: the repository ships no SRDF, planning group or kinematics plugin for TIAGo Pro. kinematics.py parses the competition URDF and solves the 8-DOF chain with damped least squares.

2 Perception

Shelf number identification

The number plates are a black digit on a white square, and the exact digit textures ship inside erc_description. That makes template matching against the ground-truth textures strictly better than OCR or a learned detector here: five classes, no dataset, no training, and no failure mode that is not visible in the mask. Dark ink is segmented in HSV, each blob is tight-cropped, normalised to 48x64 and scored by intersection-over-union against every template. A digit is accepted at IoU 0.55 and must agree on three consecutive frames before it is published.

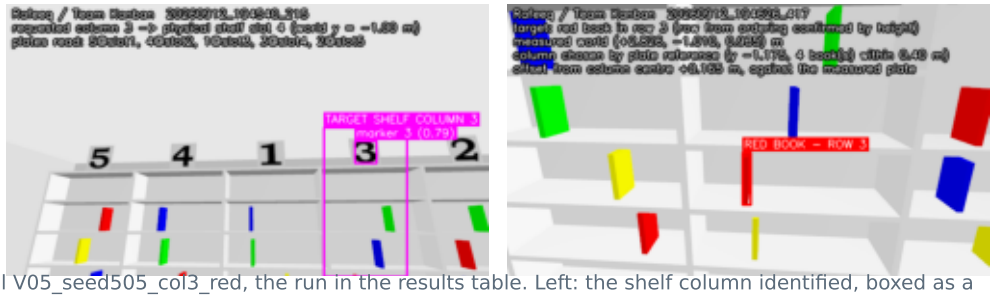
Mapping a digit to a physical column is done by measurement, not assumption: each plate is deprojected through the depth image into the world and snapped to the nearest column centre. Plates, unlike books, are not jittered - measured at exactly y = +2.0, +1.0, 0.0, -1.0, -2.0. If depth or TF is unavailable for some plates and all five are in frame, their left-to-right image order is used instead, and the fallback is logged.

Book colour, row and position

Books are unlit, fully saturated primaries presented spine-on as a 0.03 x 0.25 m rectangle, so a tight HSV band plus a 1:8 aspect test rejects almost every false positive. The row comes from two independent methods that

cross-check each other: the vertical order of the four books in a column, and the measured world height snapped to the nearest row. A disagreement is reported rather than silently resolved.

The output that matters is the measured 3D position. If depth or TF is unavailable the mode fails rather than publishing a column centre - a row number alone cannot be grasped. The column filter refuses to fall back to "all books in the image": every column holds all four colours, so such a fallback would happily return a same-colour book a metre away in a neighbouring column.



Written live by trial V05_seed505_col3_red, the run in the results table. Left: the shelf column identified, boxed as a column, with the number plate that identified it. Right: the target book inside that column, its row, and its measured offset from the column centre. These are the two images the rubric scores; the solution writes them to erc_images/ on every run.

Measured accuracy across the recorded trials

Quantity	Result
Trials recorded	10 (5 development, 5 validation)
Target column read off the number plates	10/10
Book row identified	10/10
Book 3D position measured	10/10
Mean |offset| of book from column centre	0.147 m
Largest |offset| handled	0.254 m

Both series of recorded trials count here. What changed between them was how the book is held, not how it is found, so detection is the one measurement they share. A trial counts as identified only if the value was published from an image, never from the launch argument. The mapping each run read is printed into its own evidence image, and the model name the simulator reports for the book that was grasped agrees with it every time - which is what makes these readings checkable rather than self-reported.

3 Navigation

The base is a mecanum holonomic drive, so x, y and yaw are independent. Rafeeq exploits that twice: it strafes along the shelf to put a jittered book squarely in front of the arm without turning away from the shelf, and it performs the entire final grasp approach as a pure base translation while the arm holds one validated posture.

Control is proportional on the odometry error, rotated into the body frame, with position tolerance 0.02 m and yaw tolerance 0.02 rad. Obstacle checking uses both 270-degree lidars binned into 36 sectors: the commanded direction is checked against the returns that actually lie in it, which a front-only test cannot do on a robot that can strafe and reverse. Stale odometry or scans stop the base rather than letting it integrate against a frozen pose. "Blocked" is reported separately from "reached": an earlier version conflated them and could believe it had arrived at a grasp pose it had never reached.

Travel time

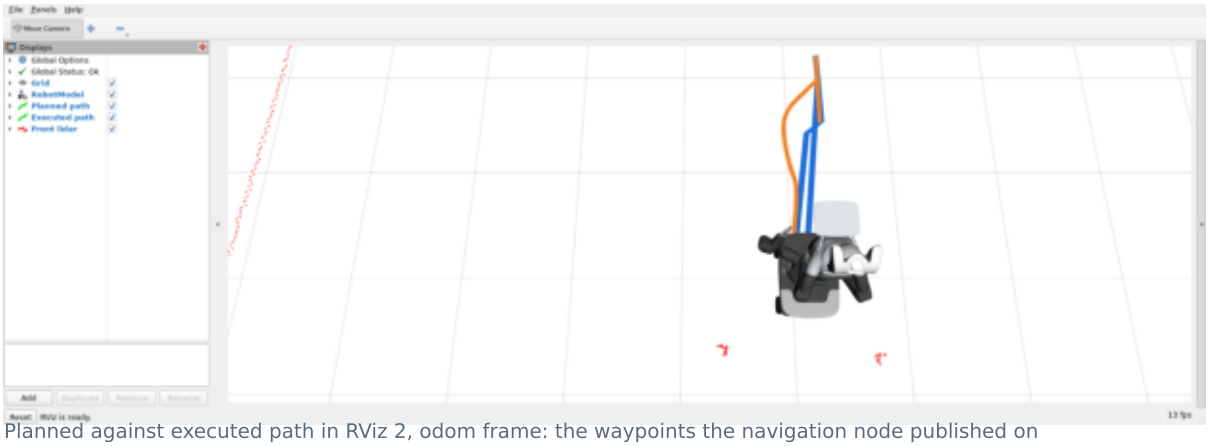
Simulated and wall-clock seconds are not the same figure - the simulator ran at 0.362 in software rendering and 0.65-0.70 on the GPU - so every time below is simulated time, which is the machine-independent one. Each leg is read out of a run's own log at build time: timed from the navigation state being entered to the arrival that ended it, and measured between the poses the base reported reaching. The take column is the recorded delivering run; the trials column is the mean of the validation trials in section 5.

Leg	Distance	Take s	Trials s (n)	m/s	Arrival error: along / across, yaw
Turn to the collection bin	turn	3.6	3.6 (5)	-	0.000 / 0.001 m, 1.1 deg
Turn to the shelf	turn	5.4	5.4 (5)	-	0.000 / 0.000 m, 1.1 deg
Start zone to the shelf column	1.41 m	8.4	8.2 (5)	0.17	0.020 / 0.001 m, 0.0 deg
Along the shelf, square to the book	0.20 m	4.4	4.9 (5)	0.04	0.019 / 0.020 m, 0.0 deg
Approach into the shelf	0.73 m	10.6	10.6 (5)	0.07	0.006 / 0.000 m, 0.0 deg
Retreat, book in the gripper	0.34 m	4.0	4.0 (5)	0.08	0.059 / 0.000 m, 0.0 deg
Carry to the collection bin	2.23 m	7.8	8.0 (2)	0.29	0.060 / 0.006 m, 0.9 deg

Whole mission, to /bin_contacts	107.8	108.8 (5)

Arrival error is the navigation node's own report of where it stopped against where it was sent, along and across the commanded heading. The approach into the shelf is deliberately the slowest leg: it is driven on the precise profile with the arm already in its grasp posture, where overshoot means striking a shelf board. Trial means cover only the trials that reached that leg, hence the count; the take ends on the delivery event, part-way through backing off, so that leg is not timed.

Three decisions account for most of the time saved, and all three buy time on the return, where a book is hanging from the pads. The bin is found from the start zone before the shelf trip, so no search leg is driven with the book held. The drive to the bin starts while the hold is still being confirmed, overlapping 1.2 s of checking with the carry. And the carry runs a higher proportional gain: at the ordinary gain the last 0.2 m is crawled at 0.05-0.12 m/s, about 2.5 s of a drive the grip beat by roughly one. Overshoot is harmless there - the aim point is short of the bin's middle.



Planned against executed path in RViz 2, odom frame: the waypoints the navigation node published on /rafeeq/nav/planned_path in blue, against the odometry the base actually drove, published on /rafeeq/nav/executed_path in orange. Grey points are the front lidar, which is what draws the shelf and the walls. Captured on run FIG_seed101_col2_red: seed 101, column 2, red book, which delivered - RViz is attached for this figure and nothing else, so it is neither the take timed above nor one of the five trials. The bow on the return is the ramped acceleration profile used while a book is held. Reproduce with tools/rafeeq_paths.rviz.

4 Manipulation

Grasping is where nearly all of the engineering went, and where the remaining blocker is. Four measured facts fix the problem almost completely.

- A book presents its spine, 0.03 m wide. Only that dimension fits between the pads, so the fingers must close along world y. This leaves the gripper roll as the single free parameter.
- The gripper body is 0.157 m long along the axis perpendicular to the fingers. With a horizontal approach that axis is vertical - and a shelf row is 0.300 m tall with a 0.250 m book already in it. The body does not fit.
- Tilting the approach rolls that body axis towards the horizontal, out of the row. Measured vertical extent of the gripper body: 0.174 m at 0 deg, 0.123 m at 45 deg, 0.060 m at 70 deg, 0.000 m at 90 deg. Whole-arm shelf intrusion reaches zero from 45 deg onwards.
- The pads separate as gap = 0.0051 + 0.777 x joint, measured by transforming the fingertip collision mesh into the grasp frame; the link origins give a misleading curve.

The sequence

```
observe -> stage 1.35 m back -> plan (roll x height x tilt) -> take posture -> base translates in   -> refine against live base pose -> close -> verify contact -> lift -> base retreats -> verify retention -> carry -> release over the bin
```

The arm takes its posture 1.35 m from the shelf, where it cannot reach the shelf while swinging, and the base carries it in. Re-solving IK at every step of an arm-driven approach instead changes the posture inside a 0.30 m gap on each one, and the controller interpolates straight through whatever lies between two postures.

Verification, and why it is not the joint position

gz_ros2_control implements a position command interface as a JointVelocityCmd computed from position error, so the finger joint is driven kinematically and reaches whatever it is told to. Commanded and achieved positions agree to five decimals whether or not a book is between the pads, and the reported effort is a constant -0.0935

N: a stall cannot be evidence of a grasp. Rafeeq reads the fingertip contact sensors instead, and requires contact to persist through the lift and a 0.42 m base retreat.

Placing the book in the bin

The bin is measured, not assumed: floor at 0.740 m, rim at 0.950 m, taken from the settled geometry rather than the spawn pose, which sits 0.35 m higher. The base drives to a pose from which the book already hangs over the bin - the gripper is a fixed offset from the base centre while the arm holds still - and the torso comes down on the way, so the descent costs no extra time and no arm motion. The gripper then opens. The implemented rubric scores that as a drop rather than a gentle placement, and the rule it uses is stated so a judge can disagree with it: gentle means the book was still held when it was lowered to within half a spine of the bin floor, and Rafeeq releases from the height it carried at, above the rim. Lowering further is arm motion, and arm motion swings a book held on a grip that lasts seconds - so this trades two placement points for the thirteen that a dropped book loses.

5 Results

The delivering run

The task has been completed end to end on the live simulator in one continuous unedited take, and the completion was reported by the organisers' own contact sensor, not by our gripper opening. Seed 101, column 2, red book. The line that ends the run:

```
OFFICIAL COMPLETION EVENT: book_col_3_row_2_red contacted the collection bin at t+107.8 s
```

Quantity	Value	Where it comes from
Delivery	t+107.8 simulated s	organiser /bin_contacts
Trial timer / video	221.3 wall s · 4:34 take	realised real-time factor 0.487
Carry, grasp to release	7.8 simulated s	against a grip good for 6-11 s
Collisions	1 (arm_right_6_link, during RETREAT)	organiser /contacts

Two clocks, and not the same clock: the mission counts simulated seconds, the video timer wall seconds. The take left no machine-readable record - the recorder ends on the delivery event, before the run writes one - so its logs, frames and checksums are at media/ and it is in no rate below. The model name is the simulator's own, read back as a check and never used to choose anything: plate 2 hangs above physical slot 3, which is the mapping the run's own evidence image records.

Five trials on the submitted code

The trial matrix re-run against the code in this repository, one record per run, written by the solution itself into results/. Same seeds, columns and colours as the development trials below, so the two sets are comparable run for run. Reproduce with tools/run_trials.sh.

trial	seed	column / colour	slot	row	offset m	grasp	delivered	coll	score	sim s
V01	101	2 / red	3	1	-0.160	yes	no	1	11.5	128
V02	202	4 / blue	2	4	-0.009	no	no	1	5.5	109
V03	303	1 / yellow	4	1	-0.139	yes	no	1	11.5	106
V04	404	5 / green	2	2	-0.253	no	no	0	6.0	100
V05	505	3 / red	4	3	+0.184	no	no	0	6.0	102

Score is a LOCAL ESTIMATE from the implemented rubric, not an official result: gentle-versus-drop and collision intent are judge decisions. Slot is the physical shelf slot the requested plate was found above; offset is the book's measured displacement from that column centre, which is what the organiser jitters. Simulated seconds.

None of the five delivered, and the cover of this report says the task was completed end to end. Both are true, and one measurement separates them: the recorded take delivered, and so did the run the navigation figure in section 3 was captured on, so delivery reproduces - it does not reproduce reliably. Every one of the five read the plates, identified the requested column and row, measured the book in three dimensions and closed on it. 3 then lost it during the extraction, before it was clear of the shelf. 2 carried it out and lost it on the way to the bin. Not one failed at the identification, and not one grasped the wrong book. Section 6 is the measurement behind that margin, and widening it is the first item of remaining work.

Rates

Metric	Value	Denominator
Target column read off the number plates	5/5	validation trials
Book row identified	5/5	validation trials
Book 3D position measured	5/5	validation trials

Grasp verified through the lift and the retreat	2/5	validation trials
Delivered to the collection bin	0/5	validation trials
Mean estimated score	8.1 / 19, against 5.9 before the fix	validation trials
Mean trial duration	109 sim s	validation trials

Development trials, before the extraction fix

The same matrix run before the work described in section 4, kept in results/ rather than deleted because it is what the problem looked like. All 5 identified the correct column, the correct row and a 3D position, and all 5 reached the grasp phase. 4 closed on the target book and then lost it, after the lift or during extraction; 1 was rejected during IK refinement before closing. None delivered. Two further runs exceeded the harness wall-clock cap and wrote no record; they are excluded rather than counted either way. The gap to the validation trials is not three points of grasping: the rubric awards shelf navigation on a successful grasp and the return on a successful delivery, so failing at the grasp costs 13 of the 19.

6 Limitations and future work

The grip is a clock

Grasp retention was the open blocker of the previous revision. It no longer blocks - the book is grasped, extracted, carried and delivered - but it is not solved, and the distinction matters. The pads sit on mimic joints whose URDF effort limit is 0.1 N m, gz_ros2_control enforces a mimic with a velocity servo capped at that limit, and nothing in the provided world offers an attach mechanism. The grip is friction against a hinge that can be pushed open: the book creeps out under its own weight and stays held, measured, for 6 to 11 seconds once it hangs free. Squeezing harder does not help - 6 mm to 16 mm of interference changed nothing.

What made delivery possible was accepting that and sizing the task to it: the bin is identified before the shelf trip, so no time is spent hunting for it with a book in hand; the drive starts while the hold is still being confirmed; and the arm holds still throughout the carry, because every arm motion swings the book. The delivering take carried it for 7.8 s against a grip good for 6 to 11. That margin is also why runs still fail: of the 5 validation trials in section 5, 0 delivered. All five closed on the correct book; 3 lost it during the extraction and 2 after carrying it clear. Under the provided simulation dynamics, widening the margin would need either more gripper-force authority than a 0.1 N m mimic hinge can offer or a different contact model in the world. Rafeeq therefore minimises transport duration and arm motion so as to work inside the retention window it measured. The measurements, the reproducer and the four remedies tried and reverted are in docs/grasp_blocker.md.

What changed during this work

Defect found	Resolution
16 states treated a timeout as success	Every transition now gates on an observation
Measured book position published, never consumed	Grasp planner consumes it; base strafes to it
Grasp poses aimed at column centres	Books jitter 0.25 m; poses solved per measurement
Gripper body fouled the row	Approach tilt search, 0 shelf contacts
Bin pose used the spawn height (1.3 m)	Measured settled geometry, rim at 0.950 m

Five of thirteen. The other eight - a home command rejected as already running, which failed a trial two seconds in; the top row unreachable on the wrong wrist roll; the arm driving through shelf boards, both grippers parked inside the shelf, trajectories timed at one guessed speed, evidence images excluded by .gitignore, the viewer losing its startup race with the robot spawn, and an unrecorded release height - are written up with their measurements in docs/.

Remaining work

In order: widen the retention margin rather than living inside it, which is the one change that would move the delivery rate. Replace the coarse advisory clearance model, a padded polyline through link origins, with swept-volume checking: it can rank IK solutions but should not veto them. Eliminate the start-up collision in which the tucked left arm brushes the table. And cover all 20 column and colour combinations rather than the recorded subset.

None of this has run on hardware, and the central finding above is specific to the simulator: a kinematic gripper on a 0.1 N m mimic hinge is not a TIAGo Pro applying a real force. A physical robot brings the reverse - odometry drift, lighting that breaks tight HSV bands, books that are not saturated primaries - so the perception needs re-grounding in real imagery before any of it transfers. Rafeeq is Arabic for companion, and the intent is a library assistant for people who cannot reach the stacks: the direction of travel, not a claim about this submission.